

Context Window – Simple Explanation

The above image explains the concept of the Context Window in Large Language Models (LLMs).

Let us understand this clearly in simple language.

Context Window

The context window is the amount of text the model can “see” and “remember” at one time during a conversation.

Everything you send — your question, previous chat history, and even the model’s earlier responses — must fit inside that window.

Think of it like the model’s short-term memory — once it’s full, older messages fall off.

Model	Context Window Limit (tokens)	Max Output Tokens (tokens)	Notes / Source
GPT-5 (OpenAI)	~ 400,000 tokens (~ 272 k input + 128 k output)	~ 128,000 tokens	Combined limit of input + output is up to ~400k tokens in API use.
Claude Sonnet / Claude 4 (Anthropic)	Up to 200,000 tokens for standard API; up to 1,000,000 tokens in extended/enterprise variants	~ 64,000 tokens for Sonnet 4 standard	Enterprise versions may have higher limits; output token caps vary.

More tokens = longer memory, but also higher cost.

1. What is a Context Window?

A context window is the amount of text a model can “see” and process at one time.

It includes:

- Your current question
- All previous conversation messages
- The assistant’s previous responses
- Any system instructions

Everything must fit inside this limit.

You can think of it like short-term memory for the model.

Once it becomes full, older messages start getting removed.

2. Why Does It Matter?

As discussed earlier:

LLMs are stateless.

They do not remember previous conversations by themselves.

So every time you send a new message:

- The application sends the entire conversation history again to the LLM.
- This full text must fit inside the context window.

If the conversation becomes very long (for example, 20–30 days in the same chat), the total number of tokens increases.

At some point:

- The context window limit is reached.
- The oldest messages are removed.
- The model stops remembering earlier parts of the conversation.

That is why sometimes it feels like the system “forgot” older messages.

3. What Happens When the Limit Is Exceeded?

If:

Input tokens + Output tokens > Context Window Limit

Then one of the following happens:

1. Oldest tokens are removed automatically, or
2. The request fails due to token overflow

This is called context truncation.

4. Tokens and Cost

Every word or part of a word is converted into tokens.

More tokens mean:

- More memory usage
- Higher computational cost
- Higher billing cost (in API usage)

So long conversations increase:

- Token consumption
- Cost

This is why people started focusing on:

Context Engineering

instead of only Prompt Engineering.

Context Engineering means:

- Carefully selecting what past information to send
- Removing unnecessary text
- Summarizing older conversations
- Managing token usage efficiently

5. Context Window vs Output Tokens

There are two important limits:

1. Context Window Limit
This controls how much total text (input + output) the model can process.
2. Maximum Output Tokens
This controls how long the model's response can be.

Example:

You can provide:

350,000 tokens of logs, documents, or code as input.

But:

The response might still be limited to 128,000 tokens.

If the total exceeds the model's limit (for example 400,000 tokens), then:

- Old tokens are dropped, or
- The request fails.

6. Key Understanding

Context Window controls:

How much the model can see.

Max Output Tokens controls:

How much the model can say.

7. Important Reminder

LLMs do not store conversation permanently.

Applications are responsible for:

- Storing chat history
- Sending previous messages again
- Managing token limits

If you keep sending the entire conversation every time:

Eventually, the context window fills up,
and older content gets removed.

Final Summary

Context Window is the memory size available during one request.

It:

- Has a fixed token limit
- Includes input + output
- Drops old messages when full
- Directly impacts cost

Understanding context window is very important when designing AI systems, especially for long conversations, document processing, and enterprise applications

Key Difference (one-liner)	
Feature	What it controls
Context Window (400k)	How much the model can see
Max Output Tokens (128k)	How much the model can say